

Curriculum Vitae

Takashi Furuya

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I am an Assistant Professor in the Faculty of Life and Medical Sciences at Doshisha University and a Visiting Scientist at the RIKEN Center for Advanced Intelligence Project.

My research interests lie at the intersection of applied mathematics and machine learning, with a particular focus on:

- Neural operators and operator learning
- Transformers and in-context learning
- Approximation theory for deep learning
- Inverse problems for partial differential equations

Contact Information

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Education and Professional Experience

- **September 2025–present:** Visiting Scientist, RIKEN Center for Advanced Intelligence Project, Computational Physics Machine Learning Team, Japan
- **April 2025–present:** Assistant Professor, Faculty of Life and Medical Sciences, Doshisha University, Japan
- **August 2022–March 2025:** Assistant Professor, Education and Research Center for Mathematical and Data Science, Shimane University, Japan
- **April 2021–July 2022:** JSPS Postdoctoral Fellow, Department of Mathematics, Hokkaido University, Japan
- **2018–2021:** Ph.D. in Mathematics, Graduate School of Mathematics, Nagoya University
- **2016–2018:** M.Sc. in Mathematics, Graduate School of Mathematics, Nagoya University
- **2012–2016:** B.Sc. in Mathematics, College of Mathematics, University of Tsukuba

Publications

Under Review

1. **T. Furuya**, D. Mis, I. Dokmanić, M. V. de Hoop, M. Lassas.
Function Graph Transformers Universally Approximate Operators between Function Spaces.
arXiv:2605.17968 (2026).
2. R. Barboni, **T. Furuya**, M. V. de Hoop, G. Peyré.
Training Infinitely Deep and Wide Transformers.
arXiv:2605.17660 (2026).
3. **T. Furuya**, Y. Korolev, T. Yaguchi.
Approximation of Maximally Monotone Operators: A Graph Convergence Perspective.
arXiv:2605.12301 (2026).
4. **T. Furuya**, R. Ozawa, Jenn-Nan Wang.
Approximation Theory of Laplacian-Based Neural Operators for Reaction-Diffusion System.
arXiv:2605.12025 (2026).

5. **T. Furuya**, A. Kratsios, D. Possamai, B. Raonić.
One Model to Solve Them All: 2BSDE Families via Neural Operators.
arXiv:2511.01125 (2025).
 6. **T. Furuya**, M. V. de Hoop, M. Lassas.
Transformers through the Lens of Support-Preserving Maps between Measures.
arXiv:2509.25611 (2025).
 7. A. Kratsios, **T. Furuya**.
Is In-Context Universality Enough? MLPs Are Also Universal In-Context.
arXiv:2502.03327 (2025).
 8. A. Kratsios, **T. Furuya**.
Simultaneously Solving FBSDEs with Neural Operators of Logarithmic Depth, Constant Width, and Sublinear Rank.
arXiv:2410.14788 (2024).
 9. A. Kratsios, **T. Furuya**, L. B. J. Antonio, M. Lassas, M. V. de Hoop.
Mixture of Experts Soften the Curse of Dimensionality in Operator Learning.
arXiv:2404.09101 (2024).
 10. **T. Furuya**, H. Kusumoto, K. Taniguchi, N. Kanno, K. Suetake.
Theoretical Error Analysis of Entropy Approximation for Gaussian Mixtures.
arXiv:2202.13059 (2022).
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Publications

1. **T. Furuya**, D. Murari, C. B. Schönlieb.
Approximation Theory for Lipschitz Continuous Transformers.
International Conference on Machine Learning (ICML), 2026.
2. A. Kratsios, B. J. Kim, **T. Furuya**.
Approximation Rates in Besov Norms and Sample-Complexity of Kolmogorov-Arnold Networks with Residual Connections.
Neural Networks, **200**, 108797 (2026).
3. **T. Furuya**, P. Z. Kow, J. N. Wang.
Consistency of the Bayes Method for the Inverse Scattering Problem with Randomly Truncated Sieve Priors.
Inverse Problems and Imaging, **21**, 222–244 (2026).
4. D. Murari, **T. Furuya**, C. B. Schönlieb.
Approximation Theory for 1-Lipschitz ResNets.
Advances in Neural Information Processing Systems (NeurIPS), **38**, 142888–142919 (2025).
5. **T. Furuya**, M. V. de Hoop, G. Peyré.
Transformers Are Universal In-Context Learners.
International Conference on Learning Representations (ICLR), 80820–80845 (2025).
6. **T. Furuya**, K. Taniguchi, S. Okuda.
Quantitative Approximation for Neural Operators in Nonlinear Parabolic Equations.
International Conference on Learning Representations (ICLR), 85528–85556 (2025).
7. **T. Furuya**, J. N. Wang.
The Bernstein–von Mises Theorem for the Inverse Scattering Problem.
Communications in Mathematical Sciences, **23**(4), 1023–1042 (2025).
8. H. S. O. Borde, **T. Furuya**, A. Kratsios, M. T. Law.
Approximation Rates and VC-Dimension Bounds for (P)ReLU MLP Mixture of Experts.
Transactions on Machine Learning Research (2025).
9. **T. Furuya**, M. Puthawala, M. Lassas, M. V. de Hoop.
Can Neural Operators Always Be Continuously Discretized?
Advances in Neural Information Processing Systems (NeurIPS), **37**, 98936–98993 (2024).

10. **T. Furuya**, S. Okuda, K. Suetake, Y. Sawada.
Convergences for Minimax Optimization Problems over Infinite-Dimensional Spaces Towards Stability in Adversarial Training.
Transactions on Machine Learning Research (2024).
11. L. B. J. Antonio, **T. Furuya**, F. Faucher, A. Kratsios, X. Tricoche, M. V. de Hoop.
Out-of-Distribution Risk Bounds for Neural Operators with Applications to the Helmholtz Equation.
Journal of Computational Physics, **513**, 113168 (2024).
12. **T. Furuya**, P. Z. Kow, J. N. Wang.
Consistency of the Bayes Method for the Inverse Scattering Problem.
Inverse Problems, **40**, 055001 (2024).
13. **T. Furuya**, M. Puthawala, M. Lassas, M. V. de Hoop.
Globally Injective and Bijective Neural Operators.
Advances in Neural Information Processing Systems (NeurIPS), **36**, 57713–57753 (2023).
14. M. V. de Hoop, **T. Furuya**, C. L. Lin, G. Nakamura, M. Vashisth.
Local Recovery of a Piecewise Constant Anisotropic Conductivity in EIT on Domains with Exposed Corners.
Inverse Problems, **39**, 025005 (2023).
15. **T. Furuya**, R. Potthast.
Inverse Medium Scattering Problems with Kalman Filter Techniques.
Inverse Problems, **38**, 095003 (2022).
16. **T. Furuya**, K. Suetake, K. Taniguchi, H. Kusumoto, R. Saiin, T. Daimon.
Spectral Pruning for Recurrent Neural Networks.
Proceedings of the 25th International Conference on Artificial Intelligence and Statistics (AISTATS), PMLR Vol. 151 (2022).
17. **T. Furuya**.
Remarks on the Factorization and Monotonicity Method for Inverse Acoustic Scattering.
Inverse Problems, **37**, 065006 (2021).
18. **T. Furuya**.
The Factorization and Monotonicity Method for the Defect in an Open Periodic Waveguide.
Inverse and Ill-Posed Problems, **28**, 783–796 (2020).
19. **T. Furuya**.
Scattering by the Local Perturbation of an Open Periodic Waveguide in the Half Plane.
Journal of Mathematical Analysis and Applications, **489**, 124149 (2020).
20. **T. Furuya**.
The Direct and Inverse Scattering Problem for the Semilinear Schrödinger Equation.
NoDEA Nonlinear Differential Equations and Applications, **27**, Article 24 (2020).
21. T. Daimon, **T. Furuya**, R. Saiin.
The Monotonicity Method for the Inverse Crack Scattering Problem.
Inverse Problems in Science and Engineering, **28**, 1570–1581 (2020).
22. **T. Furuya**.
A Modification of the Factorization Method for Scatterers with Different Physical Properties.
Mathematical Methods in the Applied Sciences, **42**, 4017–4030 (2019).
23. H. Chihara, **T. Furuya**, T. Koshikawa.
Hermite Expansions of Some Tempered Distributions.
Journal of Pseudo-Differential Operators and Applications, **9**, 105–124 (2017).

Funding and Fellowships

Competitive Research Funding

PI: Principal Investigator; Co-I: Co-Investigator.

Royal Society ISPF International Collaboration Awards 2025

Co-Investigator (Co-I)

Project: *Operator learning for gradient flows with applications in imaging*

Principal Investigator: Yury Korolev

Grant No.: ICA/R2/252033

Period: March 2026 – March 2029

JSPS KAKENHI Grant-in-Aid for Early-Career Scientists

Principal Investigator (PI)

Project: *A study of operator learning for inverse problems in partial differential equations*

Grant No.: 24K16949

Period: April 2024 – March 2027

Amount: JPY 3,900,000

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Sumitomo Foundation Grant for Basic Science Research 2023

Principal Investigator (PI)

Project: *Approximate solutions for ill-posed inverse problems of partial differential equations using deep learning*

Period: December 2023 – November 2024

Amount: JPY 1,000,000

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Yamanashi Prefectural Grant for Young Researchers

Principal Investigator (PI)

Project: *Reconstruction methods for inverse problems in partial differential equations using deep learning*

Period: August 2022 – March 2023

Amount: JPY 1,000,000

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International Research Collaboration

JSPS–NSF International Research Exchange Program

Host researcher: Prof. Michael A. Puthawala (South Dakota State University)

Host institution: Doshisha University, Japan

Period: January – February 2026

Fellowships

JSPS Postdoctoral Fellowship (PD)

Grant No.: 21J00119

Institution: Hokkaido University

Period: April 2021 – March 2023

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JSPS Research Fellowship for Young Scientists (DC2)

Institution: Nagoya University

Period: April 2019 – March 2021

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Tobitate! Study Abroad JAPAN

Host institution: Karlsruhe Institute of Technology, Germany

Period: October 2018 – February 2019

[Program page](#)